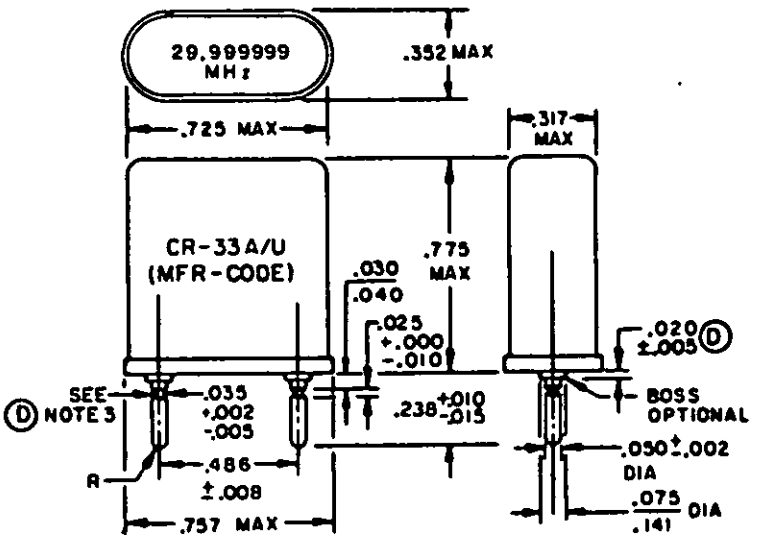
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-33A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 10 to 30 MHz; third overtone; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
Ⓓ 3. The pin undercut may be omitted.

Ⓓ denotes changes.

FSC 5955

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:19Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (See MIL-H-10056/2)
Dimensions, marking, and configuration	See figure
Frequency range	10 to 30 MHz, inclusive
Capacitance, shunt	12 pF, maximum
Frequency tolerance	±50 PPM
Equivalent resistance	See table II
Antiresonance, load capacitance	32 ±0.5 pF
Mode of oscillation	Third mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.5 ±0.5 mW
Calibration values:	
Resistance	25 ±1 ohms
Resistor voltage drop	0.25 volt
Temperature range:	
Operating	-55° to +105°C, inclusive
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
(D) Thermal shock:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	± 10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
(D) Low temperature storage:	
Resistance	See table II
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 201 of MIL-STD-202.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:19Z
Check the source to verify that this is the current version before use.

TABLE II. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
10 to 11	65
11+ to 12	60
12+ to 13	55
13+ to 14	50
14+ to 15	45
15+ to 16	41
16+ to 17	38
17+ to 18	36
18+ to 18.5	34
18.5+ to 19	32
19+ to 19.5	30
19.5+ to 20	28
20+ to 20.5	26
20.5+ to 21	25
21+ to 21.5	24
21.5+ to 22	23
22+ to 22.5	22
22.5+ to 23	21
23+ to 23.5	20
23.5+ to 24	19
24+ to 24.5	18
24.5+ to 25	17
25.5+ to 30	15

NOTES:

Ⓓ 1. The socket designed to mate with crystal holder HC-6/U is part no. M12883/04-01 (see MIL-S-12883/4). For Navy use, other sockets listed in MIL-STD-242 may be used.

2. This crystal unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0408-7)

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:19Z
Check the source to verify that this is the current version before use.